

Abhinav Hiroji

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SUMMARY

Passionate and detail-oriented **Computer Science undergraduate** specializing in **Data Science, AI, and Software Engineering**. Strong foundation in **Python, C++, and SQL** with a track record of building and deploying **data-driven applications** that combine analytical insight and backend engineering. Experienced in **deep learning (LSTM)**, and developing data pipelines for real-time prediction systems. Demonstrated ability to translate raw datasets into meaningful insights and deliver high-performance models and dashboards that support research and practical implementation.

EDUCATION

Jain University, Faculty of Engineering and Technology Bangalore, Karnataka Bachelor of Technology in Computer Science and Engineering | Oct 2022 – Jul 2026

TECHNICAL SKILLS

Languages: Python, C++, SQL, HTML, CSS, JavaScript
Frameworks/Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras, Flask, Streamlit
Concepts: Machine Learning, Deep Learning, Time Series Forecasting, Model Evaluation, Data Preprocessing, OOP
Tools: Power BI, Excel, Jupyter Notebook, VS Code, Git
Databases: MySQL

PROJECTS

- **Air Quality Index (AQI) Prediction using LSTM | TensorFlow, Keras, Flask** Jan 2024 – Apr 2024
 - Developed a multivariate **LSTM-based model** to predict Air Quality Index (AQI) using pollutants such as PM2.5, PM10, NO₂, and CO.
 - Preprocessed time-series datasets with Pandas and NumPy, applying normalization, feature scaling, and correlation analysis.
 - Achieved **92% prediction accuracy** through parameter tuning and dropout regularization in TensorFlow.
 - Integrated live AQI data through the **Gemini API**, automating daily data ingestion for real-time monitoring.
 - Deployed a **Flask-based dashboard** to visualize pollutant trends, alert levels, and health risk categories.
 - Optimized data pipeline efficiency, reducing preprocessing time by **30%** compared to baseline implementation.
- **Smart Home Automation System | Arduino, IoT Sensors, C++** Aug 2023 – Oct 2023
 - Designed a cost-effective **IoT system** that automates home appliances using temperature, light, and motion sensors.
 - Used Arduino Uno with LDR, DHT11, and PIR sensors to monitor environmental changes and trigger device responses.
 - Integrated **ESP8266 Wi-Fi module** for remote operation, allowing users to control appliances via mobile interface.
 - Implemented energy-saving algorithms that improved household power efficiency by approximately **25%**.
 - Programmed device logic in C++ ensuring safety fallback mechanisms and non-blocking interrupt handling.

COURSEWORK & CERTIFICATIONS

- Data Analytics Job Simulation – **Deloitte (Forage)** (Feb 2024): Completed data-driven business analytics project focusing on KPI visualization and insight reporting.
- Machine Learning & Artificial Intelligence – Academic Coursework (2023–2024): Developed regression, classification, and clustering projects using Python.
- Operating Systems, Database Management Systems, Computer Networks, and Data Structures & Algorithms – Core CSE subjects completed with distinction.